A Focused Survey on Patient Simulation with Large Language Models

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***Abstract*—The rapid advancement of generative artificial intelligence technologies, particularly large language models such as ChatGPT, has opened new avenues in healthcare education. One of the most promising applications is the simulation of virtual patients to enhance clinical training. This study aims to critically analyze and compare ten recent peer-reviewed research articles that explore the integration of ChatGPT-based virtual or standardized patients into medical education. The selected studies vary in methodology, scope, and application, ranging from history taking and empathy training to gamified learning environments, emotion recognition, and structured simulation platforms. A qualitative synthesis was conducted, focusing on implementation strategies, learner engagement, user satisfaction, and educational outcomes. The review reveals that large language model-powered virtual patients show high potential in improving communication skills, diagnostic reasoning, and learner autonomy. Several studies report positive user feedback and notable cost and scalability advantages compared to traditional simulation methods. In addition, novel training techniques targeting the emotional awareness of artificial intelligence agents are emerging as an important direction for future development. However, limitations such as response accuracy, emotional realism, and the absence of non-verbal communication remain key challenges. Most authors agree that large language models should complement rather than replace human interaction in simulation-based learning. This paper contributes to a deeper understanding of how generative artificial intelligence can be effectively used in medical education, providing valuable insights for educators, curriculum designers, and technology developers seeking to integrate artificial intelligence tools into training programs.**

Keywords*— Large Language Model; Virtual Patients; Healthcare Education; Generative Artificial Intelligence; Empathy; Emotion Recognition; Medical Simulation component

I. Introduction

The integration of artificial intelligence (AI) into medical education has gained rapid momentum in recent years, driven by the increasing capabilities of large language models (LLMs) such as ChatGPT. These generative AI tools are reshaping how educators design, deliver, and evaluate training in clinical settings.

Among the most innovative applications is the use of ChatGPT-powered virtual patients (VPs) and standardized patients (SPs), which offer learners safe, accessible, and cost-effective platforms to develop clinical reasoning, communication, and diagnostic skills [1, 2].

Traditional clinical education methods often rely on human actors or manikins to simulate patient encounters. While effective, these approaches are resource-intensive and constrained by logistical limitations, including personnel training, scalability, and cost [3, 4].

To address these challenges, researchers have begun exploring AI-based simulations that leverage the natural language understanding and generative capabilities of LLMs. This paradigm shift enables scalable, personalized, and repeatable training scenarios for students at all levels of medical education [5, 6].

Early studies show promising results regarding the realism, usability, and educational value of these AI-powered simulations. Vaughn et al. demonstrated that ChatGPT-generated scenarios could support faculty in developing credible and diverse case-based exercises, though limitations related to missing clinical detail and factual inaccuracies remain [4].

Similarly, Liu et al. [5] and Holderried et al. [2] confirmed that ChatGPT could simulate patient dialogues effectively, scoring high in realism and diagnostic plausibility when evaluated by medical experts. However, concerns persist regarding authenticity, especially in emotionally nuanced interactions and empathy-related contexts [7].

Another critical dimension of these tools is their impact on learner autonomy and skill development. Aster et al. found that students using ChatGPT for history-taking tasks experienced a heightened sense of autonomy and self-directed learning [7].

Meanwhile, the Korean medicine training platform developed by Ryu et al. emphasized structured competency development in specific domains such as suicide risk assessment, suggesting that these tools can be tailored for niche educational contexts [3].

Furthermore, the open-access platform described by Thesen et al. underscores the importance of transparency, scalability, and educator control in deploying generative AI tools for training purposes [8].

Despite their advantages, these systems are not without drawbacks. Issues such as occasional hallucination of facts, language biases, and lack of standardized pedagogical feedback limit their standalone use in high-stakes education [1, 5].

Additionally, most current implementations remain experimental or supplementary, highlighting the need for rigorous evaluation and integration into formal curricula [9].

This paper aims to provide a comprehensive literature-based analysis of ten recent studies exploring the use of ChatGPT and generative AI technologies as VPs in healthcare education. By comparing their methodologies, implementation strategies, evaluation metrics, and reported outcomes, the evolving role of AI in clinical training is illuminated, and both opportunities and limitations in this rapidly expanding field are identified.

II. LITERATURE REVIEW

The application of LLMs, particularly ChatGPT, in healthcare education has attracted considerable attention in recent years. This review evaluates ten studies published between 2021 and 2025 that explore the use of ChatGPT-powered VPs and SPs for clinical training. These studies vary in implementation methods, target learners, interaction style, and evaluation metrics.

Holderried et al. [2] designed a web-based chatbot using GPT-3.5-turbo to simulate a patient role in history-taking scenarios. Their findings indicate high realism and plausibility in responses, with 94% of chatbot outputs matching predefined scripts. Similarly, Liu et al. [5] implemented ChatGPT in ten clinical scenarios evaluated by senior physicians, confirming the model's ability to replicate realistic patient behavior.

Several studies explored structured or scripted VP systems. Aster et al. [7] evaluated the role of ChatGPT in simulated empathetic dialogue, noting improvements in learner autonomy but identifying limitations in emotional depth and non-verbal feedback. Pears et al. [6] introduced a gamified platform that integrates ChatGPT with video technologies, highlighting enhanced engagement but noting the lack of clinical depth due to the demo-based nature of the system.

Ryu et al. [3] proposed a simulation platform for Korean medicine education using a prompt-tuned GPT-4o. Their system achieved high student satisfaction, particularly in niche domains such as suicide risk assessment. Vaughn et al. [4] demonstrated a scenario generation tool using GPT-3.5 prompts to assist educators in case design. Though beneficial for rapid prototyping, expert revision was necessary to ensure accuracy.

In terms of deployment, Thesen et al. [8] developed an open-access web app using GPT-4o with human-authored case constraints. Their platform emphasized flexibility, multilingual access, and customization. Cologlu et al. [1] investigated intern physicians' interaction with an AI-powered VPs in clinical simulations, reporting high realism but identifying language glitches and lack of visual input.

Additional studies addressed backend architectures. Hsieh et al. [10] integrated multiple LLMs (ChatGPT, PaLM, ERNIE-4, Mistral) into a simulation engine with retrieval-augmented generation (RAG). This platform achieved an 85% match with human evaluation and improved realism in SP behavior.

Lastly, Tavarnesi et al. [9] employed manual prompt-based tools to design flexible cases for educators, offering speed and simplicity at the expense of accuracy and hallucination risk.

Overall, the literature highlights the rapid innovation and diversity of approaches in AI-powered medical training. These tools generally improve accessibility, scalability, and engagement, but often fall short in clinical depth, empathy modeling, and standardized assessment protocols.

III. METHODOLOGY

This paper employs a structured comparative literature review methodology to explore and synthesize recent research on the use of ChatGPT and related LLMs as VPs or SPs in healthcare education. This methodological approach allows for a rigorous, multi-dimensional examination of diverse implementations across different educational, clinical, and technological contexts. In particular, it enables a systematic mapping of how generative AI technologies are currently being designed, deployed, and evaluated in simulation-based medical training.

Structured literature reviews are particularly effective when comparing studies that apply varying tools and pedagogical strategies, as they enable the researcher to identify thematic patterns and methodological trends without relying solely on quantitative synthesis [9, 10]. This approach also facilitates a critical assessment of the pedagogical, technical, and evaluative dimensions of each study, ensuring that both strengths and limitations are explicitly addressed.

The reviewed literature consists of ten peer-reviewed articles published between 2021 and 2025. These studies were selected based on specific inclusion criteria: first, each study involved the use of ChatGPT or a comparable LLM such as GPT-3.5, GPT-4, or GPT-4o; second, the focus of the study was on medical or healthcare education, specifically in the simulation of patient communication, clinical reasoning, or diagnostic training; and third, the article included an evaluative component, such as expert reviews, learner feedback, or performance metrics.

All studies were published in English and represent a range of educational settings and clinical specialties, including general internal medicine [2], trauma [9], mental health [7], and Korean medicine [3]. The inclusion of diverse medical domains was intentional, as it allows the synthesis to capture both broad trends in AI-powered simulation and specialized applications tailored to niche training needs.

Following a full-text analysis of each study, a set of common analytical categories was inductively developed to ensure consistency and depth across the comparison. These categories include the type of simulation tool (e.g., chatbot, gamified simulation, prompt-based generator), the version and configuration of the LLM used (e.g., GPT-3.5, GPT-4, RAG-based architectures, LangChain), the target user group (e.g., medical students, interns, nursing educators), and the number and diversity of simulated clinical cases.

In addition to technical specifications, the studies were also coded for the pedagogical design principles applied, such as scenario complexity, adaptability to learner input, and the presence or absence of feedback mechanisms.

Furthermore, they were examined for reported advantages such as accessibility, realism, and learner autonomy [4, 5, 6], as well as for limitations including hallucination of facts, limited emotional depth, and lack of non-verbal communication [1, 7]. This dual focus on both benefits and challenges provides a balanced perspective on the educational potential of AI-powered simulation tools.

The extracted data were compiled into a comparative synthesis table (Table I), which enables side-by-side analysis of implementation strategies, evaluation methods, and outcomes. For example, Holderried et al. [2] evaluated ChatGPT's ability to generate realistic differential diagnoses across thirty clinical cases, while Aster et al. [7] focused on the model's performance in replicating empathic responses during history-taking scenarios. Ryu et al. [3] developed a structured simulation training platform for Korean medicine students, specifically targeting suicide risk assessment, illustrating how LLMs can be adapted to culturally specific educational frameworks. Similarly, Thesen et al. [8] presented an open-access generative AI tool built on GPT-4o that allowed educators to design and control clinical simulations, prioritizing transparency and adaptability. Tavarnesi et al. [9], on the other hand, used audiovisual storytelling and gamification to enhance engagement with AI-generated VPs, reflecting an emerging trend toward immersive, multimedia-enhanced simulation environments.

By analyzing these studies comparatively, it became possible to identify several emerging clusters in implementation, such as scripted versus open-ended simulations, faculty-controlled versus autonomous systems, and static versus adaptive feedback mechanisms. These patterns offer insights into how various design choices impact educational outcomes, learner engagement, and system scalability. For instance, scripted simulations tend to provide higher factual accuracy but lower flexibility, whereas open-ended models support richer learner interaction at the cost of consistency.

The thematic synthesis was guided by principles outlined in previous educational technology reviews, allowing for a rigorous yet flexible approach to data interpretation [9, 10].

While this methodology allowed for a rich, multi-dimensional comparison, certain limitations must be acknowledged. Differences in study design, inconsistent reporting of evaluation metrics, and a lack of transparency regarding AI configuration in some articles posed challenges to uniform analysis [4, 8].

Additionally, most implementations were still in pilot or early development stages, limiting the generalizability of their findings.

Another limitation concerns the absence of long-term outcome evaluations, as none of the reviewed studies provided longitudinal data on skill retention or transfer to clinical practice.

Nevertheless, the chosen methodology proved effective in mapping the current landscape of LLM-powered patient simulation tools in medical education and identifying critical success factors, educational potentials, and research gaps, laying the groundwork for more comprehensive future studies.

IV. CONCLUSION

This structured comparative review synthesized evidence from ten peer-reviewed studies investigating the application of ChatGPT and related LLMs as VPs and SPs in medical education.

By systematically comparing implementation methods, underlying model architectures, medical domains, participant demographics, and reported learning outcomes, this review established a detailed cross-study perspective on the state of AI-driven clinical simulation.

The analysis confirms that LLM-powered patient simulations can substantially enhance accessibility, adaptability, and learner autonomy, providing cost-effective, on-demand opportunities for clinical reasoning and communication training. Notably, implementations leveraging advanced prompting techniques, RAG, domain-specific fine-tuning, and interactive feedback loops demonstrated higher levels of contextual accuracy, clinical realism, and diagnostic plausibility.

Despite these promising developments, recurring limitations persist across the reviewed literature. Emotional depth remains limited, with AI systems often struggling to convey nuanced empathy or replicate the complexity of non-verbal communication.

Additionally, factual inaccuracies, or "hallucinations", occur with varying frequency, necessitating vigilant expert oversight in high-stakes scenarios. While some studies achieved successful integration of AI, based simulations into structured curricula and specialized training, such as Korean medicine, trauma management, and mental health assessment, others remained exploratory, lacking robust evaluation protocols or long-term outcome measurement. This disparity highlights the urgent need for standardized assessment frameworks, validated performance rubrics, and cross-institutional benchmarking to ensure educational reliability and clinical safety.

Looking forward, the continued evolution of multimodal LLMs, capable of processing and generating not only text but also speech, images, and potentially biometric data, presents an opportunity to address current limitations in empathy modeling and non-verbal interaction.

Incorporating adaptive feedback mechanisms, scenario branching logic, and integration with advanced clinical data simulation tools could further increase pedagogical value. Equally important are ethical and operational considerations, including bias mitigation, transparency in AI decision-making, data privacy safeguards, and educator training for effective integration of generative AI tools.

TABLE I. COMPARISON OF STUDIES

Title	Year	Tool Type	LLM Used	Diseases Covered	Participants	Advantages	Disadvantages
[7]	2025	Virtual Patient – text chatbot (empathy/history)	ChatGPT (GPT-3.5)	1 domain (Cardiology)	3rd-year medical students	Learner autonomy; empathic opportunities; repeatable practice	No non-verbal cues; variable empathy depth; text-only
[8]	2025	Open-access web app (educator-authored cases; formative feedback)	GPT-4o (OpenAI)	Multiple	NS	Open-access; multilingual; educator control; scalable	Feedback depth limited by rubric; case quality depends on uploads
[2]	2024	Simulated Patient chatbot (structured history-taking)	gpt-3.5-turbo (OpenAI)	1 case	Medical students	High plausibility (97.9%); 94.4% script-grounded answers; positive usability	Single case; occasional implausible/socially desirable answers; no non-verbal cues
[3]	2024	Simulation platform (CPX-style)	ChatGPT API (version NS)	1 domain (Suicide risk assessment)	KM students	Competency-focused; high satisfaction; standardized practice	Authenticity skepticism; niche scope; needs wider validation
[4]	2024	Scenario generation assistant (educator-facing)	ChatGPT (likely GPT-3.5)	5 scenarios	Peer reviewers	Rapid drafting; diverse cases; reduces authoring workload	Missing clinical details; accuracy requires expert curation
[6]	2024	Gamified platform (workshop)	ChatGPT-4 (OpenAI API)	Multiple	Healthcare learners	High engagement; cost-effective; supports remote/visual learning	Prototype scope; limited clinical depth; modest evaluation sample
[9]	2024	Virtual Patient system (interactive audiovisual + NLP)	NS (generative/NLP components; model not specified)	Multiple	Medical students	High realism via actor video; dialogic practice; end-of-visit feedback	Accuracy depends on NLP/generative layer; adaptability limited by authoring
[10]	2024	Standardized Patient engine (multi-LLM + RAG)	ChatGPT, PaLM, ERNIE-4, Mistral (ensemble)	Multiple	Students + expert evaluators	Higher factual consistency; improved realism via hybridization	Complexity/latency; compute cost; integration overhead
[5]	2023	Simulated Standardized Patient (text chat)	ChatGPT-3.5-turbo (OpenAI)	10 cases	Senior physicians	Rapid deployment; colloquial answers; high plausibility	Role-flip in long dialogs; repetitive phrasing; no non-verbal signals
[1]	2021	Virtual Patient (early AI VP prototype)	Non-LLM (classical NLP/rule-based)	Multiple	Medical trainees	Early low-cost VP; scalable beyond human SP constraints	Less natural language; narrower dialogue than LLM-based systems

In conclusion, while ChatGPT-powered VP/SP systems may not entirely replace traditional clinical education methods in the immediate future, they are likely to become a powerful and indispensable complement, especially in expanding access to realistic, personalized, and repeatable training experiences. With continued technological advancement and pedagogical refinement, these systems hold strong potential to shape the future of medical education in a positive and transformative way.

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